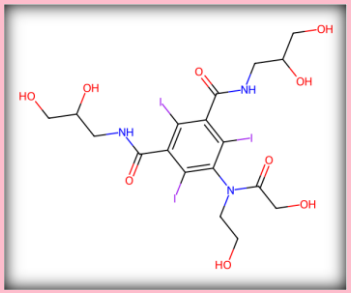
Multimodal Inputs



Molecular Graphs

OCC(O)CNC(=O)C1C(I)C(C(=O)NCC(O)CO)C(I)C(N(CCO)C(=O)CO)C1I

SMILES Strings

Fingerprints & Descriptors

ModFuse-KAN

KAN-GNN Encoder

Transformer Encoder

FP-Descriptor Encoder

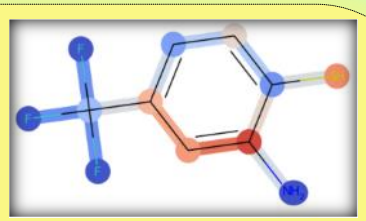
Cross-Modal Attention Fusion

Binary Molecular Property Prediction

Downstream Applications

Case Study & Interpretability

Atomic Shapley value visualization, verifying functional group toxicity links



SR-HSE Typical Sample

Early-stage drug screening

Toxicity safety assessment

Small molecule optimization

